Lab Project Documentation: Build a Game Settings Panel

1. Project Overview

This lab project involved building a custom UI settings card for a game using HTML5 and CSS3. The objective was to structure dynamic form controls and style standard HTML checkbox inputs into custom UI elements with hover/checked states and transitions.

2. Implementation Summary

HTML Structure (index.html)

- **Semantic Document Setup:** Included HTML5 document declaration (<!DOCTYPE html>), standard metadata tags for viewport rendering, and linked an external CSS stylesheet (styles.css).
- **Card Container:** Wrapped the UI in a container div with the class **.settings-card.**
- **Form Controls:** Created labeled settings options (Sound Effects, Background Music, Hard Mode, Haptic Feedback) using inline <label> elements wrapping standard <input type="checkbox" /> controls.

CSS Styling (styles.css)

- **Page & Layout:** Set the viewport height (100vh) and background color (#f0f0f0) on the body tag. Applied dynamic card dimensions using max-width: 250px, centering with margin: auto, rounded borders (10px), and drop shadow using box-shadow: 0 2px 6px rgba(0, 0, 0, 0.2).
- **Label Formatting:** Styled <label> items as block elements (display: block) with vertical margins and interactive pointer cursors (cursor: pointer).
- **Custom Input Styling:** Reset native browser checkbox styling via appearance: none. Applied explicit dimensions (20px x 20px), custom borders, background colors, and a 0.3s transition timing for smooth visual feedback.
- **State Pseudo-classes:** Targeted state changes using **input[type='checkbox']:checked** to mutate background and border colors dynamically when activated.

3. Key Concepts Learned & Practical Skills Obtained

1. **Custom Form Control Design:**
   - Mastered overriding browser defaults using **appearance: none** to build custom theme-consistent inputs from scratch without relying on external libraries.
2. **CSS Pseudo-Classes & Attribute Selectors:**
   - Utilized target-specific attribute selectors (input[type='checkbox']) and user interaction state selectors (:checked) to manage UI states declaratively.
3. **UI Layout & Alignment Techniques:**
   - Applied vertical-align: middle and margin spacing to ensure label text and interactive input boxes align horizontally.
   - Implemented card component styling using fixed max-widths, CSS box-shadow, and viewport positioning.

4. **CSS Transitions:**
   - Applied state changes (transition: all 0.3s) to produce interactive UI feedback during active toggles.

HTML CODE:





CSS CODE:







PREVIEW:

